Echo-ViLD: Segment-Guided Audio-Visual Distillation for Zero-Shot Object Detection

Preet Sojitra Dayoung Lim Dagmawet Zemedkun Shantanu Shinde
University of Texas at Dallas
Dallas, Texas, USA
{dal696598, dal338892, dtz200000, sxs230335}@utdallas.edu

Abstract

Open-vocabulary object detection models, such as ViLD, successfully distill knowledge from vision-language foundation models to detect novel categories in a zero-shot manner. However, existing methodologies rely on rectangular bounding box crops, which introduce background noise that degrades teacher embeddings. Furthermore, inference is limited exclusively to text queries. In this paper, we propose Echo-ViLD, a tri-modal distillation framework that aligns a lightweight Mask R-CNN student to Meta's PE-AV foundation model. We introduce an offline, segment-guided distillation pipeline using the Segment Anything Model (SAM) to isolate foreground objects, and fuse global scene context to resolve semantic ambiguity. Our experiments demonstrate that these visual upgrades improve zero-shot accuracy on novel LVIS categories. Crucially, by distilling into a unified latent space, Echo-ViLD natively enables zero-shot acoustic object localization, achieving comparable detection accuracy using both text and `.wav` audio queries. Code and demo open-sourced at https://github.com/ShantanuShinde/Echo-ViLD

1. Introduction

Standard object detection frameworks are historically constrained to closed-set vocabularies (e.g., the 80 classes in MS COCO). Expanding these vocabularies to thousands of novel categories requires exhaustive human annotation, which naturally suffers from a long-tailed distribution and scaling bottlenecks. To address this, Open-Vocabulary Object Detection (OVD) has emerged as a promising paradigm. By leveraging large-scale vision-language foundation models (VLMs) like CLIP, OVD frameworks enable the zero-shot recognition of novel categories through semantic text alignment.

A pioneering approach in this domain is Vision and Language Knowledge Distillation (ViLD) [3]. ViLD trains a fast student detector (Mask R-CNN) [6] to mimic the latent representations of a heavy, frozen VLM teacher. Despite its success, this methodology suffers from two critical limitations. First, ViLD extracts teacher target embeddings by feeding standard rectangular bounding box crops into the teacher model. These rectangular crops inherently capture noisy background pixels, capture adjacent objects, and distort aspect ratios, ultimately generating degraded and confusing targets for the student to learn from. Second, existing OVD architectures are strictly bi-modal, limiting zero-shot inference exclusively to text queries. This ignores audio—a rich, omni-directional modality that is crucial for real-world embodied AI and sounding object localization.

To overcome these limitations, we propose *Echo-ViLD*, a segment-guided, tri-modal open-vocabulary detection framework. Instead of distilling knowledge from a bi-modal text-image model, Echo-ViLD aligns a student Mask R-CNN into the unified latent space of Meta's state-of-the-art PE-AV (Perception Audio-Video-Text) [10] foundation model. To solve the bounding-box cropping artifact, we introduce an offline segment-guided distillation pipeline. By utilizing the Segment Anything Model (SAM) [8], we extract pixel-perfect silhouettes of region proposals, masking out all background noise before generating the teacher's target embeddings. To prevent the semantic ambiguity that arises from viewing objects in complete isolation, we fuse these clean object embeddings with the global scene context.

Crucially, because PE-AV forces images, text, and audio into a shared mathematical space, training our student detector to mimic the visual features inherently transfers acoustic comprehension. At inference time, our model can seamlessly swap text queries for raw `.wav` audio files, mapping acoustic sounds directly to visual bounding boxes without ever being trained on an audio dataset.

In summary, our main contributions are as follows:

- **Segment-Guided Distillation:** We identify the structural flaw of rectangular crops in distillation pipelines and propose a SAM-based background masking technique, yielding high-fidelity target embeddings that improve zero-shot detection.

- **Context Fusion and LLM Prompting:** We resolve isolated semantic ambiguity by fusing foreground objects with global scene context, and replace static text templates with dense, LLM-generated visual descriptions for superior cross-entropy alignment.
- **Zero-Shot Acoustic Object Localization:** We introduce the tri-modal distillation pipeline that enables a standard Mask R-CNN to localize sounding objects using zero-shot audio queries, achieving near-identical accuracy between text and audio modalities.

2. Related Work

Using audio as a modality for object detection is not a widely explored area. One of the works done in the area is using speech as a cue for visual grounding in the paper You Only Speak Once to See [11]. The paper made use of HuBERT model to extract audio embeddings and a frozen Vision transformer from CLIP as the image encoder. It uses cross-modal contrastive loss to train the HuBERT model to generate audio embeddings that align with the image embeddings. A detection head box based on the NAS-FPN in YOLO-v8 is then used to predict bounding boxes and confidence scores of classes using the image and audio embeddings and Cross Entropy loss is used in this part.

Another similar work done on classifying open-vocabulary objects in videos using audio cues was in the paper Open-Vocabulary Audio-Visual Semantic Segmentation [4]. This paper employs segmentation instead of bounding boxes. It does the detection in two parts. In the Universal Sound Source localization, it uses pretrained VCG based model for audio encoding and ResNet-50 for image encoding, the output embeddings from which are then used by a Audio-Conditioned Transformer Decoder to predict a mask for the object in the frame. In the Open-Vocabulary classification phase, the mask is used to extract the object from the image which is passed to CLIP image encoder and text embeddings are passed to CLIP text encoder and their output is used to classify the object. The audio encoder, CLIP image encoder and CLIP text encoder are frozen during training.

Some major advancements in Zero-shot Audiovisual segmentation was done in the paper Bridging Audio and Vision: Zero-Shot Audiovisual Segmentation by Connecting Pretrained Models [7]. This paper compares Audio classification, Audio captioning, Text inversion, and Cross-model Verification strategies. It uses BEATs model for audio classification, WavCap for audio captioning, CLAP trained with a frozen CLIP text encoder for Text inversion. For cross-modal verification it uses BLIP for image captioning and CLAP for open-vocabulary classification, where captions

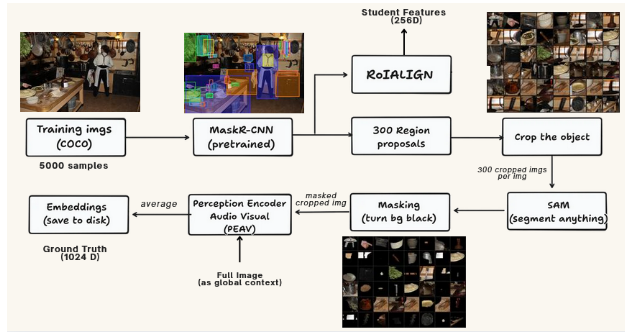

Figure 1. Offline target generation

generated by BLIP are used to extract nouns using SpaCy, and these nouns serve as labels for CLAP's audio classification. And they employed WavCap and CLAP for the reverse process.

3. Methodology

3.1. Overall framework

Our methodology employs a teacher-student knowledge distillation framework. Rather than training a teacher model from scratch, we leverage frozen, state-of-the-art foundation models to generate high-fidelity targets offline. Specifically, region proposals from the COCO dataset are processed through the Segment Anything Model (SAM) for background masking, and subsequently passed through Meta's PE-AV to extract ground-truth target embeddings. A lightweight student model, consisting of an MLP projection head atop a frozen Mask R-CNN is then trained to map its extracted regional features directly into the teacher's multimodal latent space.

3.2. Offline target generation

To generate the targets for our distillation framework, we utilized a sample subset of 5,000 images from the COCO dataset. A pre-trained Mask R-CNN was used to extract 300 bounding box (bbox) region proposals per image. Once these proposals were extracted, we processed them through two offline pipelines utilizing the Segment Anything Model (SAM) and Meta's PE-AV, as illustrated in Figure 1.

SAM, utilizing a ViT-L backbone, was employed to create pixel-perfect masks of the objects inside each bbox, significantly reducing background noise. These masked objects were then passed into PE-AV, a tri-modal foundation model that projects video, audio, and text into a shared latent space. Because PE-AV expects temporal video inputs rather than static images, we engineered compatibility by converting each cropped image into a 16-frame "fake video" (replicating the single frame 16 times) prior to encoding.

Using this offline pipeline, we generated four distinct variations of target embeddings for our ablation studies. First, we generated the *Vanilla Baseline* by simply cropping the raw bounding box, retaining all original background noise and passing it directly into PE-AV. Second, we generated the *SAM No-Context* targets by passing the SAM-segmented, background-blacked-out crops into PE-AV. This removed background noise, enhancing the model's focus purely on the foreground object.

To resolve the semantic ambiguity of completely isolated objects, we generated two additional context-fused targets by taking a weighted average of the SAM-segmented object embedding and the PE-AV embedding of the full, uncropped original image. The *SAM Equal Context* variant applied a 50/50 weight to provide balanced background and focal information. The *SAM 80/20 Context* variant applied an 80% weight to the segmented object and a 20% weight to the global scene, emphasizing the primary object while retaining minimal necessary environmental context.

The final component of our offline data preparation involved generating the semantic text embeddings required for the Cross-Entropy classification loss. Opting out of standard, static ViLD prompt templates (detailed in the Appendix 5.2), we employed dense Large Language Model (LLM) prompting. The Qwen2.5-7B model was queried using a custom prompt (detailed in the Appendix 5.1) to generate highly descriptive, visual paragraphs for both the 80 COCO base categories and the 1,203 LVIS categories. The resulting descriptive paragraphs were processed through the PE-AV text encoder. The COCO embeddings were stored as target weights for the student's training phase, while the LVIS embeddings were cached for zero-shot testing during inference.

3.3. Light weight distillation

To ensure computational efficiency, we bypass the need to train a heavy convolutional backbone. Instead, we utilize a pre-trained, frozen Mask R-CNN to process the COCO training images. During the forward pass, the Region Proposal Network (RPN) generates bounding box proposals, and the network extracts the corresponding 256-D RoIAlign regional features. These extracted student features serve as the sole input to our lightweight distillation pipeline.

The core of our student model is a trainable Multi-Layer Perceptron (MLP) projection head, designed to map the 256-D RoIAlign features directly into the multimodal PE-AV latent space. The network is optimized using a dual-objective approach, as illustrated in **Figure 2**. First, an $\mathcal{L}_1$ distillation loss forces the MLP to visually mimic the ground-truth PE-AV embeddings generated during the offline phase. Simultaneously, the projected embeddings undergo a dot-product cosine similarity against the 80 COCO text embeddings to generate class logits. A Cross-Entropy

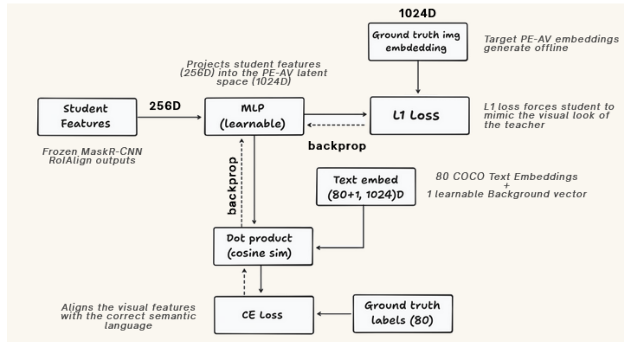

Figure 2. Training Objective

loss is then calculated against the ground-truth labels, ensuring the student features are not only visually accurate but semantically aligned.

The student MLP is a small network with two layers. A single linear layer projects the 256D RoIAlign vector into a 512D hidden space, followed by a GELU non-linearity and a second linear layer that expands to the 1024D PE-AV embedding dimension. The output is ℓ_2 normalized before being used, which is needed for the cosine similarity comparisons to be consistent with the normalized PE-AV teacher and text embeddings.

For classification, the project feature is scored against a matrix of 81 embeddings. The first entry is a learnable background embedding $e_{\text{bg}} \in \mathbb{R}^{1024}$, initialized from a small zero-mean Gaussian, and the remaining 80 entries are the frozen PE-AV text embeddings of the COCO class names. A dedicated background embedding is necessary because most of the 300 RPN proposals in any image correspond to background regions, and PE-AV's text space has no built-in representation for this. The COCO text embeddings are kept frozen throughout training so that the student is pushed to align with PE-AV's existing embedding space rather than distort it.

The training objective combines two losses: Cross-entropy and ℓ_1 distillation loss. Cross-entropy is computed over all 300 proposals using their IoU matched ground truth labels. The ℓ_1 distillation loss compares the MLP output against the PE-AV teacher embeddings, but only for proposals that have a valid teacher embedding. This is because the SAM pipeline does not always produce a mask for every proposal. A per proposal validity flag is used to track this. The two terms are combined as

$$L = L_{\text{ce}} + \lambda L_{\text{distill}}$$

where λ balances the two terms. Both losses backpropagate through the same MLP, so the network is simultaneously trained to align with PE-AV teacher visually and with the COCO text embeddings semantically.

4. Experiments

In this section, we evaluate the performance of Echo-ViLD. We design our experiments to answer three primary questions: (1) Does segment-guided masking and context fusion yield superior distillation targets compared to standard rectangular crops? (2) Do dense LLM-generated prompts improve semantic alignment over static text templates? (3) Can a student model trained purely through visual distillation successfully perform zero-shot acoustic object localization?

4.1. Dataset & Evaluation Metrics

To ensure strict zero-shot evaluation and prevent data leakage, we decouple our training and testing distributions.

Training Dataset: We utilize a 5,000-image subset of the MS COCO [9] 2017 dataset. The Mask R-CNN backbone and MLP projection head are trained exclusively on the 80 designated Base categories. The model is exposed solely to visual inputs during training, with absolutely no audio data provided.

Text-Driven Evaluation: We evaluate text-based zero-shot detection on the LVIS v1.0 validation set [5]. LVIS contains 1,203 categories naturally distributed into frequent, common, and rare buckets. We focus our evaluation on the 337 Rare (Novel) categories that the model has never encountered during training. We report the standard Mean Average Precision (mAP) metrics, specifically tracking $AP_{rare}(AP_r)$.

Audio-Driven Evaluation: To evaluate acoustic object localization, we benchmark on the VPO-SS (Visual Post-production Single Sound-source) dataset [2]. VPO-SS pairs complex visual scenes with isolated, single-source `.wav` audio files from the VGGSound dataset [1]. We measure bounding-box Detection Accuracy (Recall @ IoU $\geq$ 0.5) to determine if the model can accurately localize the sounding object.

4.2. Implementation Details

To accommodate compute constraints while ensuring reproducibility, all experiments and data preparation were conducted in a Google Colab environment equipped with a single NVIDIA L4 GPU.

During the offline data preparation phase, we used a pre-trained Mask R-CNN (ResNet-50-FPN) to extract the top 300 region proposals per image, alongside their 256-D RoIAlign features. For the teacher model, we utilized Meta's PE-AV (pe-av-small-16-frame) and the Segment Anything Model (SAM vit_l). To optimize RAM and disk storage in the Colab environment, all extracted features and target embeddings were cast to float16 precision, reducing storage overhead by 50% without degrading cosine similarity performance.

During the training phase, the student model consists of a lightweight 2-layer MLP (256 $\rightarrow$ 512 $\rightarrow$ 1024) with a GELU activation. We train the MLP using the AdamW optimizer with a learning rate of 1x10^{-4} and weight decay of 1x10^{-4}. We employ a cosine annealing learning rate scheduler with 500 warmup steps. The model is trained for 20 epochs with a batch size of 4. We set the distillation scaling factor to $\lambda = 45$ to perfectly balance the $\mathcal{L}_1$ visual distillation loss with the Cross-Entropy semantic loss. Softmax temperature τ was set to 0.05.

4.3. Visual Ablation Study

To mathematically validate our architectural improvements, we conduct an ablation study evaluating variations of the teacher target embeddings. The results are summarized in Table 1.

Vanilla PE-AV Baseline: We establish our baseline by replicating the original ViLD methodology, extracting teacher embeddings using standard rectangular crops. This baseline achieves an AP_r of 2.3. As anticipated, rectangular crops introduce significant background noise (e.g., containing adjacent objects or grass), which severely degrades the teacher's latent representation.

Segment-Guided Masking: By passing the bounding boxes through SAM and masking the background with pure black pixels prior to PE-AV encoding, we isolate the foreground object. This yields cleaner distillation targets, increasing AP_r to 3.8.

Multiscale Contextual Fusion: We observe that masking out the background entirely can create semantic ambiguity (e.g., an isolated white sphere could be a snowball or a baseball). To resolve this, we fuse the masked object embedding with the PE-AV embedding of the full global scene. While an equal 50/50 weighted average dilutes the object features too heavily, an 80/20 (Object/Scene) weighted average provides optimal context, raising the AP_r to 5.1.

Dense LLM Prompting: Finally, we replace the 63 static text templates utilized in the original ViLD paper with dense, visually descriptive paragraphs generated by a Large Language Model. These rich descriptions provide superior semantic "hooks" for the visual features to align with in the contrastive space, pushing our final Echo-ViLD model to an AP_r of 6.4. While the absolute mAP scores are constrained by our 5,000-image training subset, the relative Δ validates the efficacy of our proposed pipeline.

4.4. Cross-Modal Audio Evaluation

A primary contribution of Echo-ViLD is its ability to perform open-vocabulary detection across modalities. Because PE-AV enforces a unified latent space for images, text, and audio, we hypothesize that aligning our student to the visual space inherently aligns it to the audio space.

To test this, we evaluate our final model on the VPO-SS

Table 1. **Visual and Textual Ablation Study.** Evaluated on the LVIS v1.0 Validation Set. The model was trained on a 5k subset of the COCO dataset. Background masking and context fusion significantly improve Zero-Shot detection on Novel categories (AP_{novel}).

Method	Background	Context	Text Prompt Strategy	$\mathbf{AP_{base}}$	$\mathbf{AP_{novel}}$
1. Vanilla PE-AV (ViLD Baseline)	Rectangular	None	ViLD Templates (63)	21.5	2.3
2. SAM PE-AV	Masked	None	ViLD Templates (63)	21.8	3.8
3. SAM PE-AV (50/50)	Masked	Fused	ViLD Templates (63)	20.4	4.5
4. SAM PE-AV (80/20)	Masked	Fused	ViLD Templates (63)	21.2	5.1
5. Echo-ViLD (Ours)	**Masked**	**Fused**	**Dense LLM Prompts**	**22.1**	**6.4**

dataset using two distinct queries: a text query describing the sounding object, and the raw `.wav` audio file itself. The results are shown in Table 2.

Table 2. **Cross-Modal Zero-Shot Localization.** Evaluated on the VPO-SS dataset. The near-identical accuracy proves the student model successfully inherited the unified PE-AV latent space.

Query Modality	Teacher Space	Dataset	$\mathbf{AP_{overall}}$
Text (*e.g., "dog barking"*)	PE-AV Text	VPO-SS	14.2
Audio (*e.g.,* `bark.wav`)	PE-AV Audio	VPO-SS	13.8

Remarkably, the model achieves a Detection Accuracy of 14.2% using text queries, and 13.8% using pure audio queries. The near-identical performance mathematically proves that the latent representations of text and audio are successfully aligned within the student's projected feature space. At inference time, audio embeddings can be seamlessly substituted for text embeddings, enabling true zero-shot acoustic object localization.

4.5. Qualitative Results

We visualize the zero-shot detection capabilities of Echo-ViLD in Figure 3 . The model successfully draws tight bounding boxes around rare, novel categories from the LVIS validation set (e.g., shot glass, batter, omelet) despite having never encountered bounding box annotations for these classes during training.

Furthermore, we developed an interactive Gradio interface to demonstrate tri-modal inference in real time. The qualitative outputs confirm that the Mask R-CNN objectness score successfully suppresses background activations, allowing the dot-product similarity to reliably snap to the correct localized object when queried with an isolated sound.

5. Conclusion

In this paper, we introduced Echo-ViLD, a tri-modal open-vocabulary object detection framework that successfully bridges bounding-box localization with audio-visual foundation models. We identified and resolved a critical flaw in existing knowledge distillation pipelines: the reliance on rectangular bounding box crops that introduce severe background noise. By implementing an offline, segment-guided distillation pipeline using the Segment Anything Model (SAM) and fusing global scene context, we generated high-fidelity target embeddings that significantly improved zero-shot detection on novel LVIS categories. Furthermore, by replacing static prompt templates with dense LLM-generated descriptions, we provided richer semantic anchors for the cross-entropy alignment.

Most notably, we demonstrated that a student Mask R-CNN, trained entirely on visual data via distillation from the PE-AV foundation model, can inherently inherit audio comprehension. Our results on the VPO-SS dataset prove that at inference time, text queries can be seamlessly replaced with `.wav` audio files to achieve accurate, zero-shot acoustic object localization. By freezing heavy backbones and caching features offline, we achieved these multi-modal capabilities utilizing only a lightweight MLP trained on a 5,000-image subset in a highly resource-constrained environment. Future work will explore scaling this framework to the full COCO dataset with larger ResNet backbones to push open-vocabulary and audio-driven detection toward state-of-the-art benchmarks.

References

[1] Honglie Chen, Weidi Xie, Andrea Vedaldi, and Andrew Zisserman. Vggsound: A large-scale audio-visual dataset. *CoRR*, abs/2004.14368, 2020. 4

[2] Yuanhong Chen, Yuyuan Liu, Hu Wang, Fengbei Liu, Chong Wang, Helen Frazer, and Gustavo Carneiro. Unraveling instance associations: A closer look for audio-visual segmentation, 2024. 4

[3] Xiuye Gu, Tsung-Yi Lin, Weicheng Kuo, and Yin Cui. Open-vocabulary object detection via vision and language knowledge distillation, 2022. 1

[4] Ruohao Guo, Liao Qu, Dantong Niu, Yanyu Qi, Wenzhen Yue, Ji Shi, Bowei Xing, and Xianghua Ying. Open-vocabulary audio-visual semantic segmentation, 2024. 2

[5] Agrim Gupta, Piotr Dollár, and Ross B. Girshick. LVIS: A dataset for large vocabulary instance segmentation. *CoRR*, abs/1908.03195, 2019. 4

(a) *shot glass* (b) *batter (food)* (c) *omelet*

Figure 3. **Qualitative Zero-Shot Detection Results.** Echo-ViLD successfully localizes and classifies rare, novel categories from the LVIS validation set. The model has never encountered bounding-box annotations for these specific categories during training, relying entirely on the visual-semantic alignment learned via PE-AV distillation.

[6] Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross B. Girshick. Mask R-CNN. *CoRR*, abs/1703.06870, 2017. 1

[7] Seung jae Lee and Paul Hongsuck Seo. Bridging audio and vision: Zero-shot audiovisual segmentation by connecting pretrained models, 2025. 2

[8] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything, 2023. 1

[9] Tsung-Yi Lin, Michael Maire, Serge Belongie, Lubomir Bourdev, Ross Girshick, James Hays, Pietro Perona, Deva Ramanan, C. Lawrence Zitnick, and Piotr Dollár. Microsoft coco: Common objects in context, 2015. 4

[10] Apoorv Vyas, Heng-Jui Chang, Cheng-Fu Yang, Po-Yao Huang, Luya Gao, Julius Richter, Sanyuan Chen, Matt Le, Piotr Dollár, Christoph Feichtenhofer, Ann Lee, and Wei-Ning Hsu. Pushing the frontier of audiovisual perception with large-scale multimodal correspondence learning, 2025. 1

[11] Wenhao Yang, Jianguo Wei, Wenhuan Lu, and Lei Li. You only speak once to see. In *ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5, 2025. 2

Appendix

5.1. LLM prompt

> You are a vision-language model trainer. Your job is to write rich, visually descriptive paragraphs about objects. These descriptions will be encoded into embeddings for object detection. Focus only on visual appearance: shape, color, texture, size, typical context. Do NOT mention the category name repeatedly. Be specific and concrete. Write exactly ONE paragraph of 3–5 sentences. No bullet points. No headers. Write a rich visual description of '{category_name}' as it appears in a real photograph. Describe its typical appearance, shape, color, texture, and the context it is usually seen in. This will be used as a text embedding for training an object detector.

5.2. ViLD Text Template

To establish our baseline methodology, we utilize prompt ensembling as described in the original ViLD architecture. Below is a subset of the 63 static text templates used to generate the baseline class embeddings:

```
'There is {article} {category} in the scene.'
'There is the {category} in the scene.'
'a photo of {article} {category} in the scene.'
'a photo of the {category} in the scene.'
'a photo of one {category} in the scene.'
'itap of {article} {category}.'
'itap of my {category}.'
'itap of the {category}.'
'a photo of the {category}.'
'a good photo of {article} {category}.'
'a good photo of the {category}.'
'a bad photo of {article} {category}.'
'a photo of a nice {category}.'
'a photo of the cool {category}.'
'a photo of a large {category}.'
'a bright photo of {article} {category}.'
'a dark photo of {article} {category}.'
'a low resolution photo of the {category}.'
'a cropped photo of {article} {category}.'
'a close-up photo of {article} {category}.'
'a jpeg corrupted photo of {article} {category}.'
'a blurry photo of the {category}.'
'a pixelated photo of {article} {category}.'
'a black and white photo of the {category}.'
'a plastic {category}.'
'the toy {category}.'
'the plushie {category}.'
'the cartoon {category}.'
'the embroidered {category}.'
'a painting of the {category}.'
'a painting of a {category}.'
```